Course 02323 Introduction to Statistics 02402 Statistics (Polytechnical Foundation)

Lecture 6: Analysis of Two Samples

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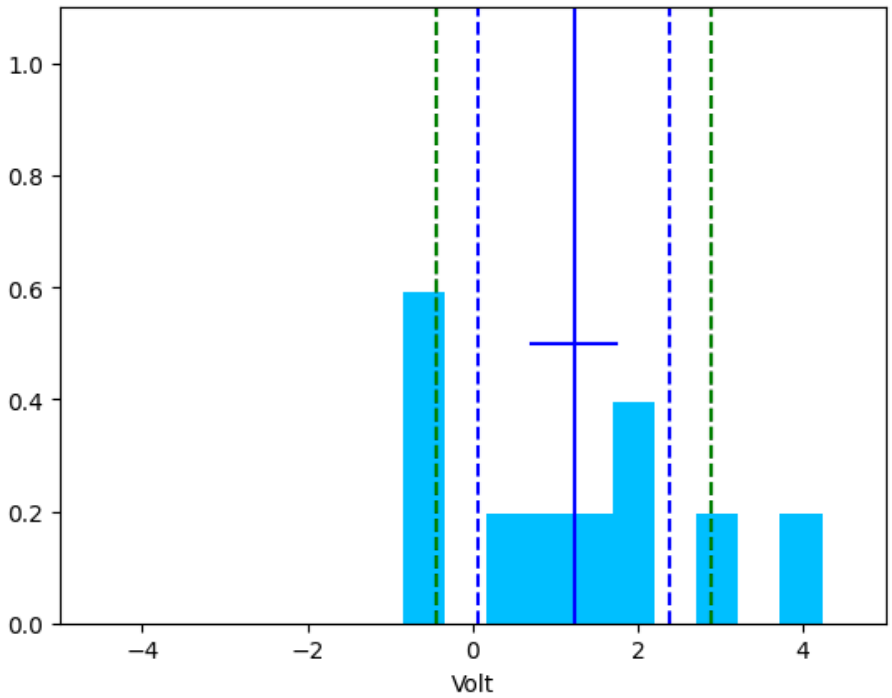
Overview

1. Summary from last week
2. (1) t-test for the difference between two samples
3. (2) t-test with pooled variance – An alternative
4. Confidence interval for the difference
5. Overlapping confidence intervals?
6. (3) t-test with two paired samples (paired t-test)
7. Assumptions and how to check them
8. Power and sample size – Experimental design
   - Precision requirements
   - Power and sample size – One sample
   - Power and sample size – Two samples

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Hypothesis testing



t-test "cook-book":

- Null hypothesis $H_0 : \mu = \mu_0 (= 0?)$
- Calculate: $\bar{x}$, s, and $s.e._{\bar{x}} = s/\sqrt{n}$
- Calculate test statistic: $t_{obs} = \frac{\bar{x} - \mu_0}{s.e._{\bar{x}}}$
- Calculate p-value: $2 \cdot P(T > |t_{obs}|)$ (lookup in t-distribution, draw a diagram)
- Compare p-value with desired α

- Is the value μ_0 *outside* the $(1-\alpha) \cdot 100\%$ confidence interval? (e.g., 95%-CI)
- Is the calculated $|t_{obs}|$ *greater* than $t_{1-\alpha/2}$? (e.g., $t_{0.975}$)
- Is the p-value *smaller* than α? (e.g., 0.05)

Yes? - then we reject the null hypothesis at significance level α.

The p-value corresponds to the significance level (α) that would just include the null hypothesis in the confidence interval.

Today



Today

- We often want to know if there is a ("significant") difference between two different things/groups/interventions.
- We have seen examples where data reflect measured differences—and we test whether this difference in means is zero.
- But what if, for example, we want to know if there is a difference between two different groups where we cannot collect data for individual differences? (the difference cannot be measured on a single statistical unit, and the groups may also be of different sizes)
- In that case, we need to look at the t-test for **two** samples.

Motivating example: Nutrition study

Difference in Energy Expenditure?

In a nutrition study, we want to investigate whether there is a difference in energy expenditure for different types of (moderately physically demanding) work.

In the study, 9 nurses from hospital A and 9 (other) nurses from hospital B had their energy expenditure measured. The measurements are shown in the following table (in the unit megajoules, MJ):

Sample from each hospital:
$n_1 = n_2 = 9$:

Hospital A	Hospital B
7.53	9.21
7.48	11.51
8.08	12.79
8.09	11.85
10.15	9.97
8.40	8.79
10.88	9.69
6.13	9.68
7.90	9.19

Python

- Go to today's Python notebook in VS Code
  - "Example: Nutrition study 1"



Kahoot!
(x1)

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Difference between two groups



What is the null hypothesis? How would you formulate it? (discuss for 2 minutes)

(board)

Difference between two groups



Considerations for the null hypothesis:

- There is no difference between groups "A" and "B"
- Do the two samples come from the *exact same* underlying distribution? (No, but both are assumed to be Normally Distributed)
- **The underlying distributions have the same mean:** $\mu_A = \mu_B$
- Does underlying distributions have the same *variance*: $\sigma_A = \sigma_B$

Example: Nutrition study

The hypothesis of no difference (in average energy expenditure) is to be tested:

$$H_0: \ \mu_A = \mu_B \quad \text{or} \quad H_0: \ \mu_A - \mu_B = 0$$

Example: Nutrition study

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Sample Means and Standard Deviations:

$$\hat{\mu}_A = \bar{x}_A = 8.293 \ (s_A = 1.428)$$
$$\hat{\mu}_B = \bar{x}_B = 10.298 \ (s_B = 1.398)$$

Example: Nutrition study

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Are the data consistent with the null hypothesis H_0?

Data: $\bar{x}_B - \bar{x}_A = 2.005$

Null hypothesis: $H_0: \ \mu_B - \mu_A = 0$

Example: Nutrition study

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Are the data consistent with the null hypothesis H_0?

Data: $\bar{x}_B - \bar{x}_A = 2.005$

Null hypothesis: $H_0: \ \mu_B - \mu_A = 0$

NEW: **p-value for the difference:**

$$p = 0.0083$$

(Calculated under the assumption that H_0 is true.)

NEW: **Confidence Interval for the Difference**:

$$2.005 \pm 1.412 = [0.59; \ 3.42]$$

Null hypothesis for a difference between two groups

What is our null hypothesis?

$$H_0: \quad \delta = \mu_X - \mu_Y = \delta_0$$

Notation: δ is the "true" difference μ_X and μ_Y are the "true" means)

Often the null hypothesis is:

$$H_0: \quad \delta = \mu_X - \mu_Y = 0$$

There is "no difference" (between the population means μ_X and μ_Y).

The difference between two means

We describe the **difference** with a single random variable:

$$D = \bar{X} - \bar{Y}$$

Mean of D:

$$\mathbf{E}[D] = \mathbf{E}[\bar{X} - \bar{Y}] = \mathbf{E}[\bar{X}] - \mathbf{E}[\bar{Y}] = \mu_X - \mu_Y$$

Variance of D:

$$\mathbf{V}[D] = \mathbf{V}[\bar{X} - \bar{Y}] = 1^2\mathbf{V}[\bar{X}] + (-1)^2\mathbf{V}[\bar{Y}] = \frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}$$

Standard deviation of D:

$$\sigma_D = \sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}}$$

Kahoot!

(x1)

Standard error for a **difference**

Standard Error for the difference:

$$\sigma_D = \sqrt{\frac{S_X^2}{n_X} + \frac{S_Y^2}{n_Y}} = \sqrt{(\sigma_{\bar{X}})^2 + (\sigma_{\bar{Y}})^2}$$

or

$$SE_d = \sqrt{(SE_{\bar{x}})^2 + (SE_{\bar{y}})^2}$$

(This is not defined in the book)

The test statistic t_{obs} for ("Welch") test with two samples

For a t-test with one sample:

We consider $T = \frac{\bar{X}-\mu}{S/\sqrt{n}}$, which follows a t-distribution with $\nu = n-1$ degrees of freedom.

From the sample data, we calculate $t_{obs} = \frac{\bar{x}-\mu_0}{s/\sqrt{n}}$, and this is used to perform the t-test.

$$t_{obs} = \frac{\bar{x}-\mu_0}{SE_{\bar{x}}}$$

For a t-test with two samples (this version is called a Welch t-test):

We consider $T = \frac{(\bar{X}-\bar{Y})-\delta}{\sqrt{S_X^2/n_X+S_Y^2/n_Y}}$, which follows a t-distribution.

From the sample data, we calculate $t_{obs} = \frac{(\bar{x}-\bar{y})-\delta_0}{\sqrt{s_X^2/n_X+s_Y^2/n_Y}}$, and this is used to perform the t-test.

$$t_{obs} = \frac{d-\delta_0}{SE_d}$$

Distribution for T ("Welch" test with two samples)

Theorem 3.50 The distribution of the (Welch) two-sample statistic

The (Welch) two-sample statistic seen as a random variable

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - \delta_0}{\sqrt{S_1^2/n_1 + S_2^2/n_2}}, \tag{3-49}$$

approximately, under the null hypothesis, follows a t-distribution with ν degrees of freedom, where

$$\nu = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}, \tag{3-50}$$

if the two population distributions are normal or if the two sample sizes are large enough.

("Welch") t-test for difference between two samples

Method 3.51 The level α two-sample t-test

1. Compute the test statistic using Equation (3-48) and ν from Equation (3-50)

$$t_{\text{obs}} = \frac{(\bar{x}_1 - \bar{x}_2) - \delta_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}} \text{ and } \nu = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1-1} + \frac{(s_2^2/n_2)^2}{n_2-1}}$$

2. Compute the evidence against the *null hypothesis*[a]

$$H_0: \quad \mu_1 - \mu_2 = \delta_0,$$

vs. the *alternative hypothesis*

$$H_1: \quad \mu_1 - \mu_2 \neq \delta_0,$$

by the

$$p\text{-value} = 2 \cdot P(T > |t_{\text{obs}}|),$$

where the t-distribution with ν degrees of freedom is used

3. If p-value $< \alpha$: we reject H_0, otherwise we accept H_0,

or

The rejection/acceptance conclusion can equivalently be based on the critical value(s) $\pm t_{1-\alpha/2}$:
if $|t_{\text{obs}}| > t_{1-\alpha/2}$ we reject H_0, otherwise we accept H_0

$$t_{obs} = \frac{d - \delta_0}{SE_d} = \frac{(\bar{x} - \bar{y}) - \delta_0}{\sqrt{s_X^2/n_X + s_Y^2/n_Y}}$$

Assumptions:

The test is valid when both samples are large (CLT), or when both samples come from normally distributed populations.

Method 3.49: The test statistic in a (Welch) t-test with two unpaired samples

Assumptions:

The test is valid when both samples are large (CLT), or when both samples come from normally distributed populations.

Calculation of the observed test statistic:

We consider the following null hypothesis about the difference in means between two *independent* and *unpaired* samples:

$$\delta = \mu_1 - \mu_2,$$

$$H_0 : \ \delta = \delta_0,$$

then the observed test statistic is:

$$t_{\text{obs}} = \frac{(\bar{x}_1 - \bar{x}_2) - \delta_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}} \quad = \frac{d - \delta_0}{s.e._d}$$

Theorem 3.50: The distribution of the (Welch) two-sample statistic

The test statistic is (approximately) t-distributed:

Under the null hypothesis, the test statistic is:

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - \delta_0}{\sqrt{S_1^2/n_1 + S_2^2/n_2}}$$

It is approximately distributed as a t-distribution with ν degrees of freedom, where

$$\nu = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$$

if the two populations are normally distributed or the sample sizes are sufficiently large.

Example: Nutrition study - Welch t-test

Perform a Welch t-test:

- Null hypothesis $H_0 : \delta = \delta_0 (= 0?)$
- Calculate: $d = \bar{x} - \bar{y}$ and $s.e._d$
- Calculate test statistic: $t_{obs} = \frac{d - \delta_0}{s.e._d}$
- Calculate p-value: $2 \cdot P(T > |t_{obs}|)$ (lookup in t-distribution)
- Compare p-value with desired α

One can use the function:

- `stats.ttest_ind(A, B, equal_var=False)` (from `Scipy.stats` package)



- Go to today's Python notebook in VS Code
  - "Example: Nutrition study 2"

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t-test with pooled variance for two unpaired samples

The *pooled* variance estimate (assuming $\sigma_1^2 = \sigma_2^2$)

Method 3.52

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}.$$

The test statistic in a t-test with pooled variance, Method 3.53

Consider the null hypothesis about the difference in means between two *independent* samples:

$$\delta = \mu_1 - \mu_2,$$
$$H_0 : \ \delta = \delta_0,$$

then the test statistic in a t-test with the pooled variance is given by:

$$t_{\text{obs}} = \frac{(\bar{x}_1 - \bar{x}_2) - \delta_0}{\sqrt{s_p^2/n_1 + s_p^2/n_2}}.$$

Theorem 3.54: The distribution of the test statistic, T ("Pooled")

Result

The test statistic in a t-test with pooled variance:

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - \delta_0}{\sqrt{S_p^2/n_1 + S_p^2/n_2}}$$

follows under the null hypothesis (and under the assumption $\sigma_1^2 = \sigma_2^2$) a t-distribution with $n_1 + n_2 - 2$ degrees of freedom, if the two populations are normally distributed.

Example: Nutrition Study - pooled t-test

Repeat the t-test, but this time under the assumption $\sigma_X^2 = \sigma_Y^2$ ("pooled" t-test)

- State the null hypothesis $H_0 : \delta = \delta_0 (= 0?)$
- Calculate: $d = \bar{x} - \bar{y}$ and SE_d – requires calculation of s_p^2
- Calculate the test statistic: $t_{obs} = \frac{d - \delta_0}{SE_d}$
- Calculate the p-value: $2 \cdot P(T > |t_{obs}|)$
  (lookup in the t-distribution with the correct ν. NOTE: ν is not the same as before)
- Compare the p-value with the desired α

You can use the function:

- `stats.ttest_ind(A, B, equal_var=True)` (from the `Scipy.stats` package)
- Go to Python notebook "Nutrition study 3" in VS Code



We always use the "Welch" version

The fewer assumptions, the better — therefore use the "Welch" version

- If $s_1^2 = s_2^2$, then the two tests are the same. If the variances are very different, the two test statistics can be significantly different. In such cases, we do not necessarily prefer the test with the pooled variance, as the assumption of equal variances may be highly questionable.
- Only if the two variances are very different, can it happen that the two tests yield very different results? If the variances seem very different, the assumption of equal variances is likely violated.
- In cases with a small sample size in at least one of the groups, the Welch version is a more "cautious" approach.
- Later in the course (when we compare several groups), however, we will assume equal variance across all groups.

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Method 3.47: Confidence interval for $\mu_1 - \mu_2$

Confidence interval for the difference in means:

For two samples $(x_1, \ldots, x_{n_1})$ and $(y_1, \ldots, y_{n_2})$, the $(1-\alpha)$-confidence interval for $\mu_1 - \mu_2$ is given by:

$$\bar{x} - \bar{y} \pm t_{1-\alpha/2} \cdot \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}},$$

where $t_{1-\alpha/2}$ is the $(1-\alpha/2)$-quantile in the t-distribution with ν degrees of freedom (given in Theorem 3-50).

$$d \pm t_{1-\alpha/2} \cdot SE_d$$

It is the same "recipe" as last time.
Be sure to calculate ν to insert as the degrees of freedom in the t-distribution (use "Welch" version.

Confidence interval for the difference between two samples

Method 3.47 The two-sample confidence interval for $\mu_1 - \mu_2$

For two samples $x_1, \ldots, x_n$ and $y_1, \ldots, y_n$ the $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is given by

$$\bar{x} - \bar{y} \pm t_{1-\alpha/2} \cdot \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}, \tag{3-45}$$

where $t_{1-\alpha/2}$ is the $(1-\alpha/2)$-quantile from the t-distribution with ν degrees of freedom given from Equation (3-50)

$$\nu = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}. \tag{3-46}$$

Example: Nutrition Study

You can also use:

- `stats.ttest_ind(A, B, equal_var=False).confidence_interval(...)`

- Go to the Python notebook"Nutrition study 4" in VS Code

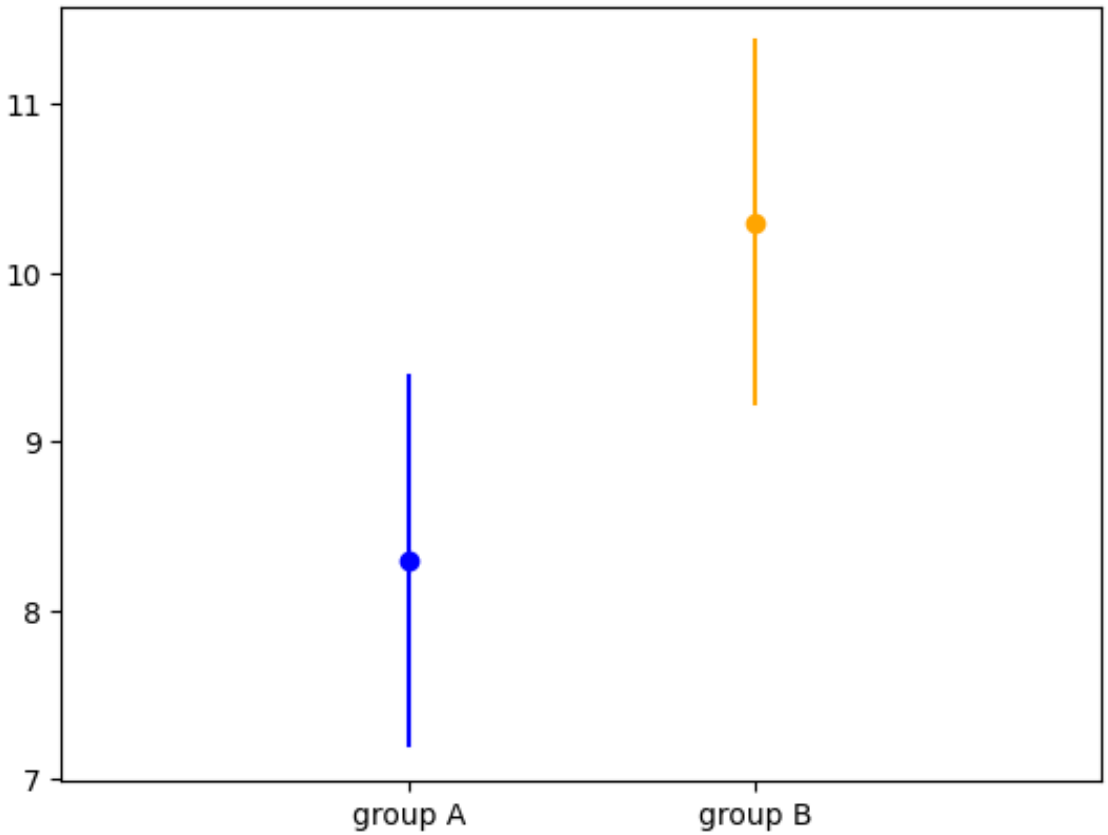


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Overlapping confidence intervals

Do you remember the overlapping confidence intervals of sample means for group A & B?

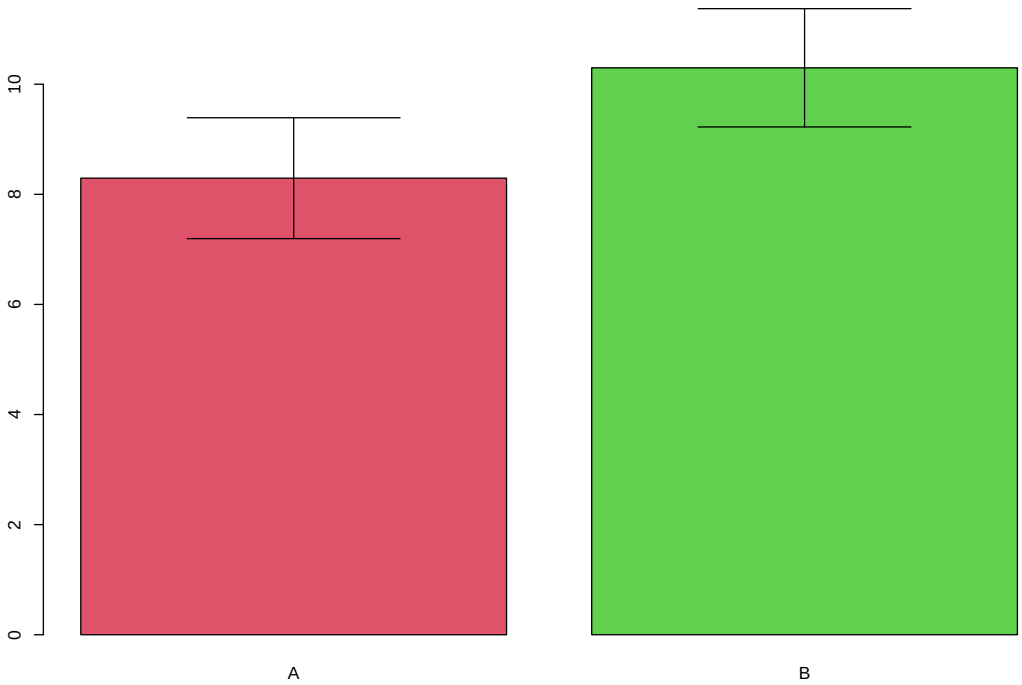


Even though the two confidence intervals overlap slightly, we concluded that there is a significant difference.

Alternative presentation from the book

Bar Charts with *error bars* are commonly seen:

A bar chart with some error bars: Below are the 95% confidence intervals for each group:



Be cautious with "overlapping confidence intervals"

When two confidence intervals DO NOT overlap: The two groups are significantly different.

When two confidence intervals overlap: No conclusion can be drawn without examining the confidence interval for the *difference* between the groups.

Remark 3.59

When interpreting two (and multi-) independent samples mean bar plots with added confidence intervals:

When two CIs do NOT overlap: The two groups are significantly different

When two CIs DO overlap: We do not know from this what the conclusion is (but then we can use the presented two-sample test method)

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Motivating example: Sleep medication

Difference in Sleep medication?

In a study, the aim is to compare two sleep aids, A and B. From 10 test subjects. The following results were obtained, indicating the extended sleep time in hours (the difference in the effect of the two drugs is noted):

Sample with $n = 10$:

Person	A	B	$D = B - A$
1	+0.7	+1.9	+1.2
2	-1.6	+0.8	+2.4
3	-0.2	+1.1	+1.3
4	-1.2	+0.1	+1.3
5	-1.0	-0.1	+0.9
6	+3.4	+4.4	+1.0
7	+3.7	+5.5	+1.8
8	+0.8	+1.6	+0.8
9	0.0	+4.6	+4.6
10	+2.0	+3.4	+1.4

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5	-1.0	-0.1	+0.9
6	+3.4	+4.4	+1.0
7	+3.7	+5.5	+1.8
8	+0.8	+1.6	+0.8
9	0.0	+4.6	+4.6
10	+2.0	+3.4	+1.4

$\bar{x} = 1.67$
$s = 1.13$

Sleep Medicine:t-test for paired data

When you have two "paired" samples, these can be *translated* into one sample with measured differences. The data are therefore analyzed using an ordinary one-sample t-test.

You can use:

- `stats.ttest_1samp(diff_data, pop_mean=0)`
  Or:
- `stats.ttest_rel(A, B)`

(_*rel* stands for *relative*, i.e., paired data)



- Go to the Python notebook "Example: Sleep medicine (paired t-test)" in VS Code

Experimental setup: Paired and independent samples

Completely Randomized (Independent Samples):

We have 20 patients, randomly assigned to two groups (typically equal numbers in each group). This means there are different (independent) patients in the two groups.

Paired Observations (Dependent Samples):

We have 10 patients, all of whom receive both treatments (typically with some time between and random order of treatments).
This means the same patients appear in both groups.

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Assumptions Behind t-Tests

The t-tests presented earlier rely on assumptions about the sampling process and the population distribution. From Theorem 3.3: $X_1, X_2, \ldots, X_n \sim N(\mu, \sigma^2)$
This implies two key assumptions:

- **Independent observations**
- **Normal distribution**

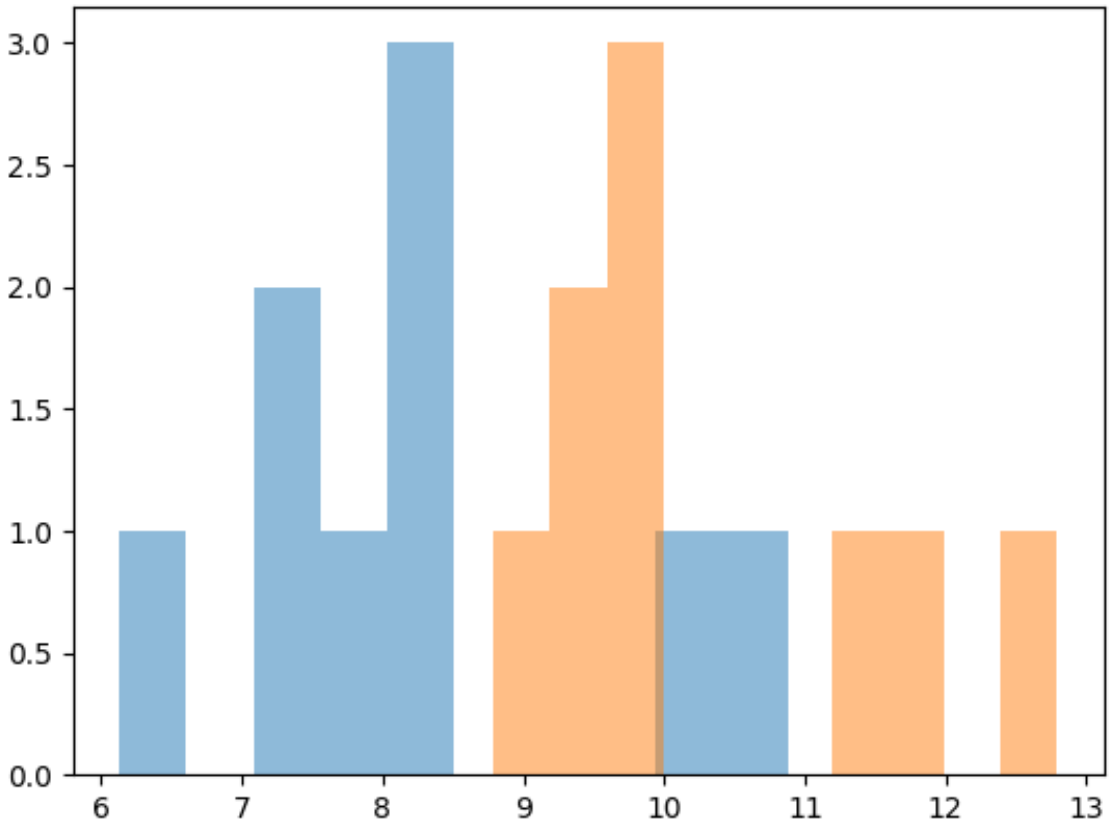
Independent observations

- Independence is mainly assessed from the **sampling procedure**.
- Example: measuring student heights — including several family members (e.g., twins) would violate independence.

Normal distribution

- The population from which the sample is drawn should be approximately normal.
- Normality can be checked **graphically** using the sample data.
- Common tools: histograms, empirical cumulative distribution function, Q–Q plots..

Normality assumptions



We revisit data on energy consumption among nurses at hospital A and hospital B.

QQ-plot

- Go to today's Python notebook in VS Code
  - "QQ-plots"

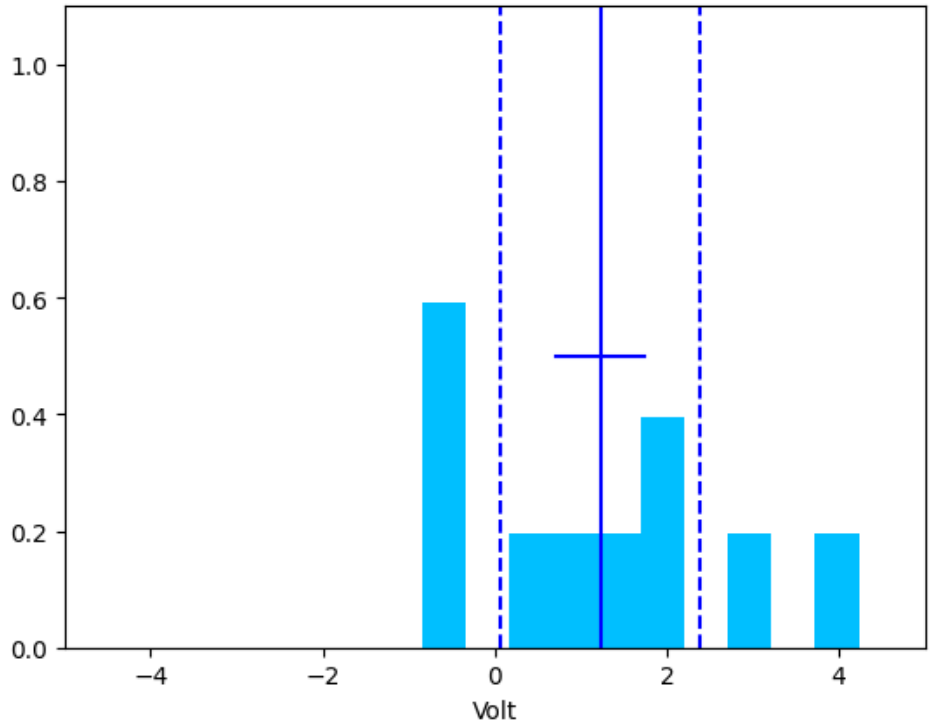


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   - Power and sample size – Two samples

Example - Light rail (letbane)

Let's revisit the Light Rail example. The data measures the voltage drop across a point where the voltage drop should be zero on average.



From the sample (n=10):

$\bar{x} = 1.23$

$s = 1.62$

Confidence interval:

$\bar{x} \pm t_{1-\alpha/2} \cdot (s/\sqrt{n}) =$

1.23 ± 1.16

$[0.07; 2.9]$

In this case, we have: $\hat{\mu} = 1.23 \pm 1.16$ volts $(\alpha = 0.05)$.
We state that the **"Margin of Error" (ME)** is 1.16 (volts).

Example - Light rail

The conclusion from the sample was that we reject the null hypothesis — thus we conclude that the voltage drop is NOT zero (on average).

But the manager is not satisfied and says we must investigate the matter better.
Our margin of error of ± 1.16 volts is too large.

We plan to take a new sample.

How can we do better than the first time?

What can we do to improve our estimate of μ?

Kahoot!
(x1)

Experimental planning with precision requirements or margin of error

Margin of Error (ME) is defined as

$$ME = t_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}$$

For sufficiently large samples, we can approximate $t_{1-\alpha/2}$ (from the t-distribution) with $z_{1-\alpha/2}$ (from the normal distribution).

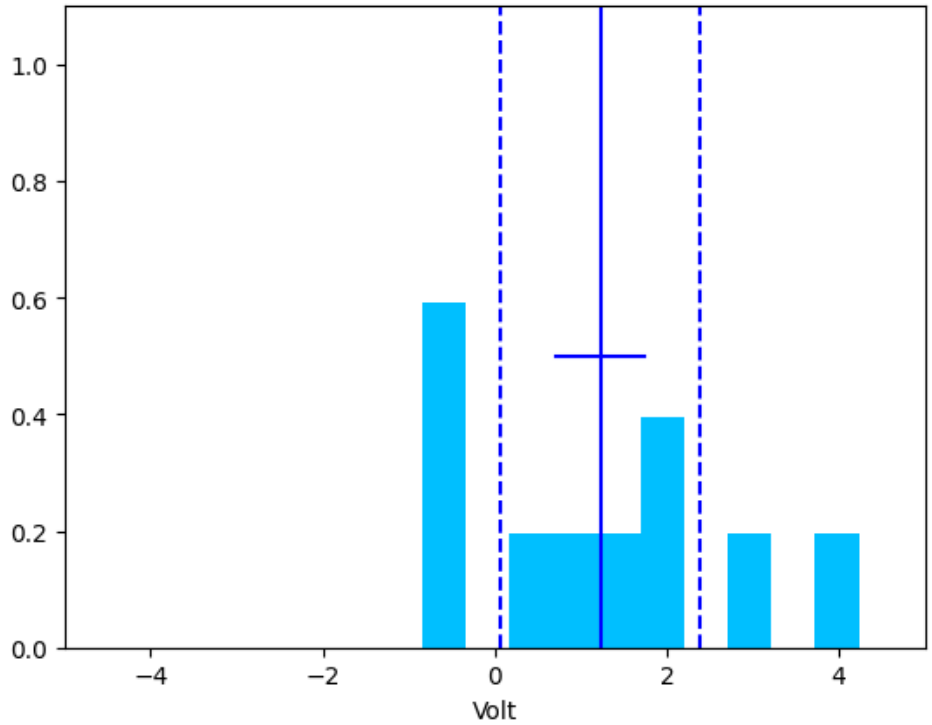
Method 3.63: Sample Size for Confidence Interval Based on a Sample

If σ is known, or estimated to be a certain value, we can calculate the required sample size n to achieve a given margin of error, with probability $1-\alpha$.

$$n = \left(\frac{z_{1-\alpha/2}\cdot\sigma}{ME}\right)^2$$

Example - Light rail ($ME = 0.5$ volts)

Let's estimate the sample size if we want ME to be at most 0.5 volts. We assume that $s = 1.62$ is a reasonable guess for the true σ.

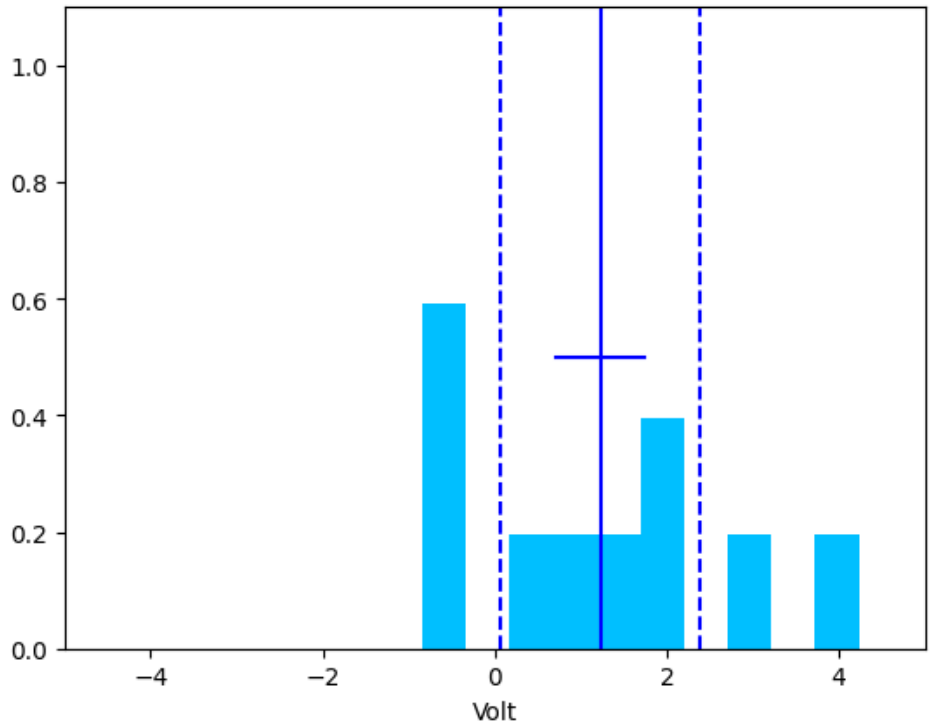


$$
\begin{aligned}
n &= \left(\frac{z_{1-\alpha/2} \cdot \sigma}{ME} \right)^2 \\
&= \left(\frac{1.96 \cdot 1.62}{0.5} \right)^2 \\
&= 40.3
\end{aligned}
$$

Since we cannot take 40.3 measurements, we round up to 41 (in the new sample, $n = 41$).

Example - Light Rail ($ME = \pm 1$ volts)

Let us estimate the sample size if we want a margin of error (ME) of at most ± 1 volt.
We assume that $s = 1.62$ is a reasonable estimate of the true σ.



$$
\begin{aligned}
n &= \left(\frac{z_{1-\alpha/2} \cdot \sigma}{ME} \right)^2 \\
&= \left(\frac{1.96 \cdot 1.62}{1} \right)^2 \\
&= 10.08
\end{aligned}
$$

Since we cannot take 10.08 measurements, we round up to 11 (in the new sample, $n = 11$).

Planning (Power and sample size)

We will now talk about experimental planning and "power".

To do this, we need to consider the two types of errors that we introduced last time:

Type I: Rejecting H_0 when H_0 is true.

$$P(\text{Type I error}) = \alpha$$

Type II: Accepting (not rejecting) H_0 when H_1 is true.

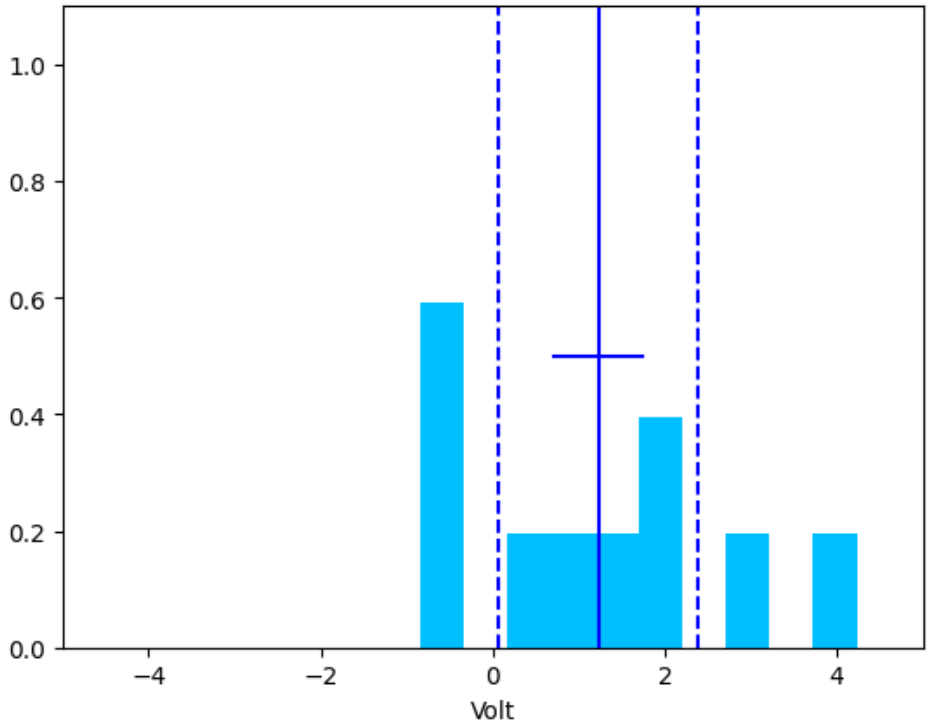
$$P(\text{Type II error}) = \beta$$

Example: Light rail

The manager is now becoming a bit more demanding. She says that if the true average voltage drop is ± 0.5 volts (or more), we want to know it.

How should we go about it?

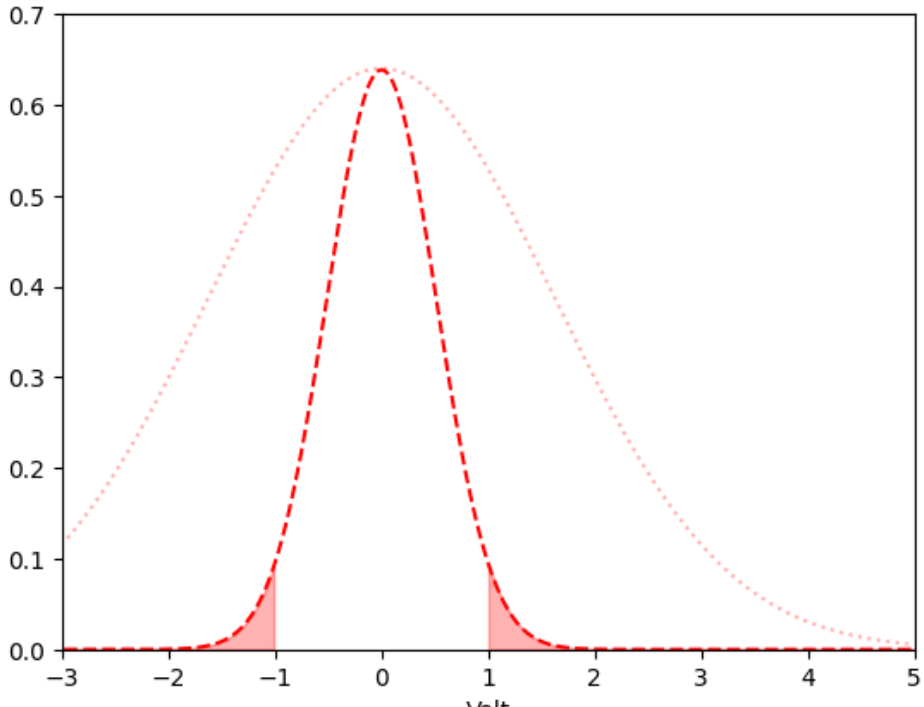
We will now go through some different **scenarios**.



Example: Light rail

Scenario 1: The null hypothesis is true

- *If* the true average voltage drop *is* zero
- and we take another round of measurements - this time with a larger n
- then the risk of incorrectly rejecting the null hypothesis will be α (e.g., 0.05 or 5%)



The values of $\bar{x}$ that we potentially accept as being "not significantly different from zero" lie in the range $0 \pm ME$. The width of this range depends on the size of the sample (n).

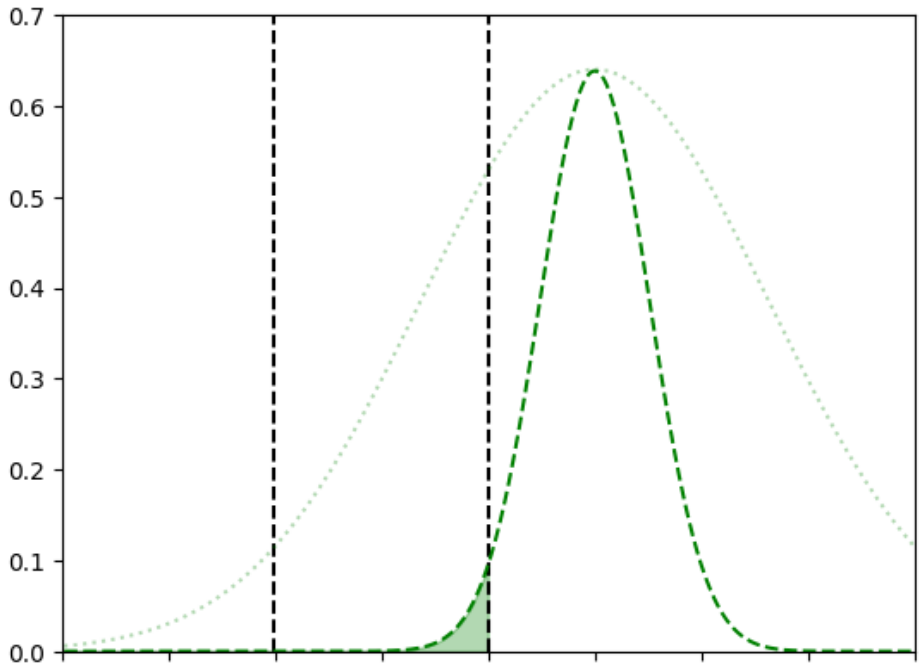
Kahoot!

(x2)

Example: Light rail

Scenario 2: The alternative hypothesis is true

- *If* the true average voltage drop is something other than zero - the true value is μ_1
- and we take another round of measurements - this time with a larger n
- What is then the risk (probability β) of accepting (not rejecting) the null hypothesis?

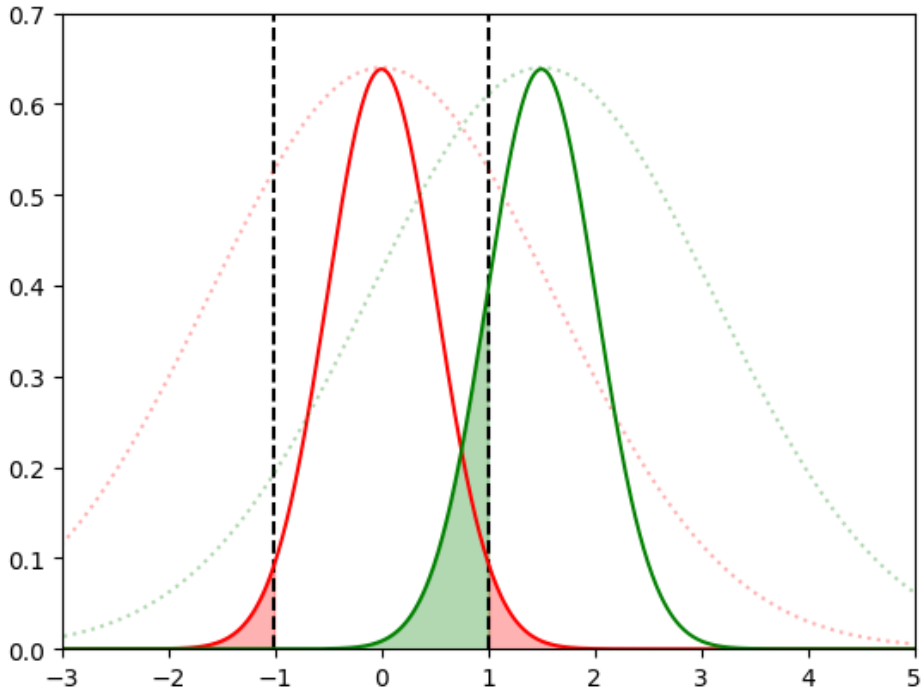


Yes, it depends on...
...how large μ_1 actually is.

Kahoot!
(x2)

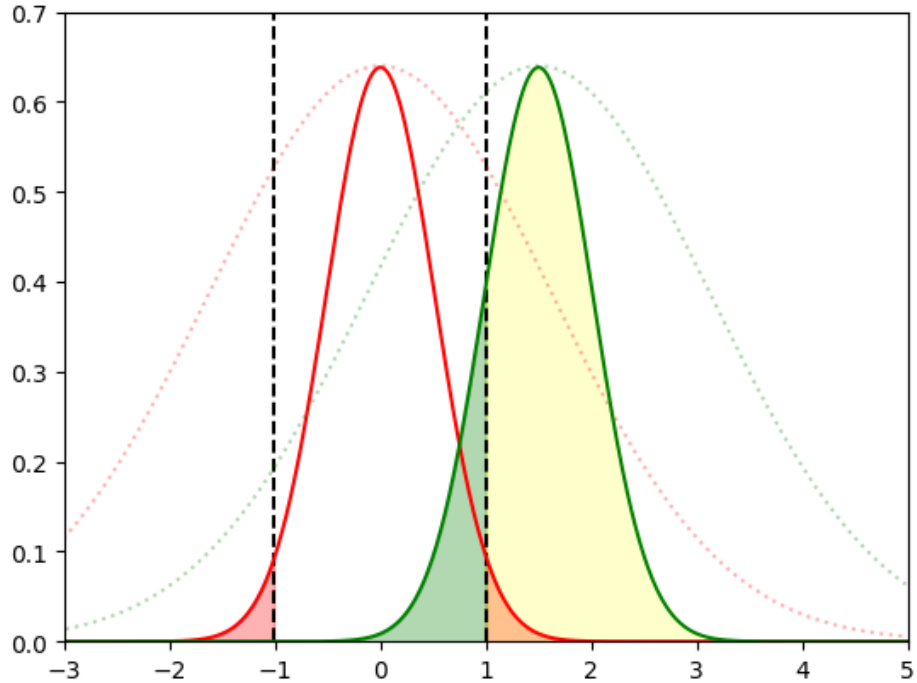
Type I and Type II errors

Red Curve: Represents the null hypothesis (H0), which assumes no effect or no difference.
Green Curve: Represents the alternative hypothesis (H1), which assumes there is an effect or difference.



α: the red area β: the green area $(1-\beta)$ is called **power** (eng: **power**)

Experimental planning: Power



- **Power** $= 1 - \beta$
- $P(\text{Type II error}) = \beta$
- Power = "Probability of correctly rejecting H_0."
- Power = "Probability of detecting a (proposed) effect."
- Typical values for desired power are 80%, 90%, etc. (something relatively large)
- In practice: Use a scenario-based approach

Kahoot! (x3)

Experimental planning: Power

If we know (or assume) four out of the five following quantities, we can find the missing one:

- Sample size, n.
- Significance level, α, at which we are testing.
- Difference in means (effect size), $\mu_0 - \mu_1$.
- Population standard deviation, σ.
- Power, $1 - \beta$.

Experimental planning: Sample size n

The Big Question: How large should n be?

We need enough observations to detect a relevant effect with high power $1-\beta$ (typically at least 80%):

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Method 3.65: Formula for Sample Size with One Sample

For a t-test with one sample, where α, β, and σ are given:

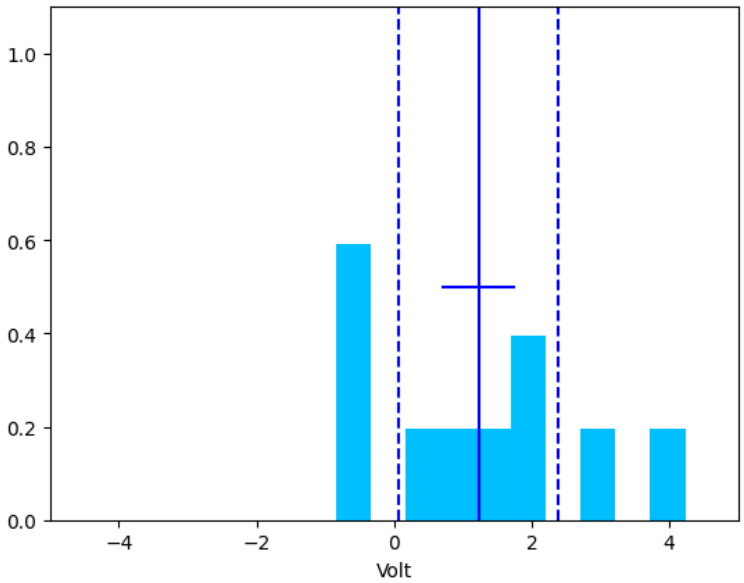
$$n = \left(\sigma \frac{z_{1-\beta} + z_{1-\alpha/2}}{\mu_0 - \mu_1} \right)^2.$$

Here, $\mu_0 - \mu_1$ is the difference in means that we want to measure, while $z_{1-\beta}$ and $z_{1-\alpha/2}$ are quantiles of the standard normal distribution.

Example: Light rail

We now tell the boss that we can design the next sample such that we will have an 80% probability of detecting a voltage drop of 0.5 volts (or more) at a significance level of $\alpha = 0.05$.

To calculate the necessary sample size, we use (as an informed guess): $\sigma = 1.62$, corresponding to the spread we observed in our previous measurements.



We find that the next sample needs to have 83 measurements:

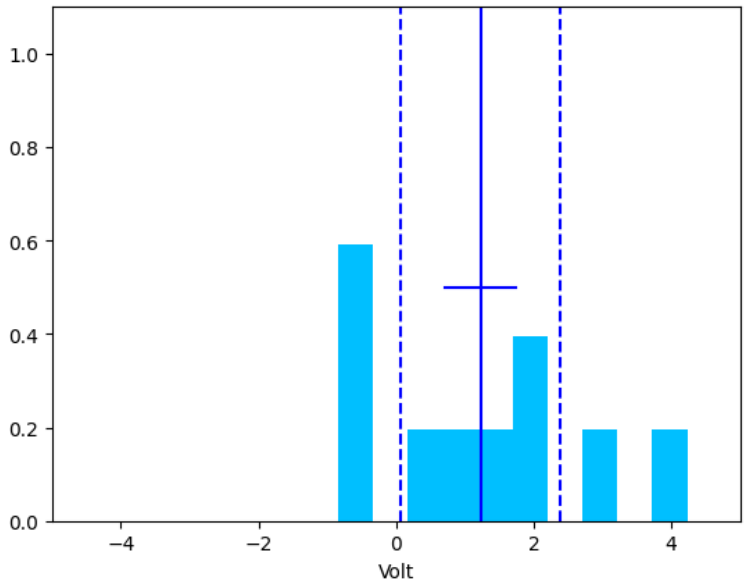
$$
\begin{aligned}
n &= \left(\sigma \frac{z_{1-\beta} + z_{1-\alpha/2}}{\mu_0 - \mu_1} \right)^2 \\
&= \left(1.62 \frac{z_{0.80} + z_{0.975}}{0.5} \right)^2 \\
&= 82.4
\end{aligned}
$$

Example: Light rail

"83 measurements are too expensive," says the boss.

"Can we settle for 60 measurements?"

Yes... - but then our experiment will have lower power, the probability of detecting a voltage drop of 0.5 volts a the given significance level.



New power calculation:

$$n = \left(\sigma \frac{z_{1-\beta} + z_{1-\alpha/2}}{\mu_0 - \mu_1} \right)^2$$

$$\Rightarrow$$

$$z_{1-\beta} = \sqrt{60 \frac{0.5^2}{1.62^2}} - z_{0.975}$$

$$= 0.43$$

Corresponding to a power of 0.67

Python

We will now use some functions from the package `statsmodels.stats.power`

(`import statsmodels.stats.power as smp`)

You can use:

- `smp.TTestPower().solve_power(effect_size=delta/sd, alpha=0.05, power=0.80)`

- Go to the Python notebook in VS Code



Experimental planning: Two samples

The big question: How large should n_1 and n_2 be?

We can generalize the experimental planning to the situation where we compare two samples. The sample sizes are now given by n_1 and n_2. We will assume that the variance (σ^2) is equal in the two populations.

(This formula is not included in the book)

Experimental planning: Two samples

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We can generalize the experimental planning to the situation where we compare two samples. The sample sizes are now given by n_1 and n_2. We will assume that the variance (σ^2) is equal in the two populations.

Formula for Sample Size with Two Samples

For a t-test with two samples, where α, β, and σ are given:

$$n_1 = (k+1)\left(\sigma \frac{z_{1-\beta} + z_{1-\alpha/2}}{\mu_1 - \mu_2}\right)^2$$

Here, $\mu_1 - \mu_2$ is the difference in means that we want to measure, while $z_{1-\beta}$ and $z_{1-\alpha/2}$ are quantiles of the standard normal distribution. Additionally, $k = n_1/n_2$ is the ratio of the two sample sizes, so we have $n_2 = n_1/k$.

(This formula is not included in the book)

Python

We will now use the function from the package `statsmodels.stats.power`

(`import statsmodels.stats.power as smp`)

You can use:

- `smp.TTestPower().solve_power(effect_size=delta/sd, alpha=0.05, power=0.80)`

- Go to the Python notebook "Example: Power calculations, 2 samples" in VS Code



Agenda



1. Summary from last week
2. (1) t-test for the difference between two samples
3. (2) t-test with pooled variance – An alternative
4. Confidence interval for the difference
5. Overlapping confidence intervals?
6. (3) t-test with two paired samples (paired t-test)
7. Assumptions and how to check them
8. Power and sample size – Experimental design
   - Precision requirements
   - Power and sample size – One sample
   - Power and sample size – Two samples

Checklist

After today, you should be able to:

- Calculate the standard error for a difference
- Perform a t-test for two samples – both Welch and pooled
- Explain the difference between the Welch and pooled t-test
- Calculate the confidence interval for a difference
- Explain when a paired t-test should be performed and be able to carry it out
- Describe what is meant by Type 1 and Type 2 errors
- Describe what is meant by statistical power
- Calculate the required sample size n, based on α, β, σ, and $\mu_0 - \mu_A$
- Python: Use functions to perform various t-tests
- Python: Use functions to calculate sample size, power, etc.